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Profile

Seasoned professional with **22 years of global experience in Artificial Intelligence (AI/GenAI), Cloud Architecture, and Automation Consulting**, specializing in building innovative, scalable, and enterprise-grade solutions for Fortune 500 clients. **With 7 years of deep expertise in AI and Generative AI**, I have successfully driven applied R&D, large-scale implementations, and AI-enabled business transformation across Healthcare, Life Sciences, Energy, Resources, and Utilities.

As AI Practice Head at TCS, I lead one of the organization's largest business units, defining and executing the AI strategy across regulated industries. My role involves building and scaling global AI practices, incubating GenAI-powered solutions in areas such as clinical decision support, sustainable energy optimization, predictive maintenance, and customer experience transformation, while establishing AI governance guardrails for safety, compliance, and ethical adoption.

A certified GSDC Generative AI Professional and AWS Certified Solutions Architect – Professional, I bring proven expertise in enterprise AI adoption, cloud-native architectures, automation frameworks, and SDLC acceleration using AI. Recognized for driving innovation across matrix-structured teams in the USA, UK, and Europe, I excel at influencing CxO-level stakeholders, fostering partnerships with hyperscalers and academia, and positioning AI as a driver of operational efficiency, compliance, and business growth.

Achievements

Pioneered Generative AI Solutions: Designed and deployed a cutting-edge **AI-powered conversational agent** for a leading healthcare provider, improving customer query resolution time by **45%** and increasing user satisfaction by **30%**.

Advanced Generative Model Development: Spearheaded the development of a novel GAN-based image synthesis model, achieving state-of-the-art performance in medical imaging diagnostics, enhancing accuracy by 20%.

Transformative AI Innovation: Led the creation of a transformer-based generative language model tailored for legal document summarization, reducing processing time by 60% and boosting productivity in client deliverables.

Implemented MLOps pipelines using tools like MLflow and SageMaker, streamlining the deployment of AI models and reducing deployment time by 30%.

Cloud Migration Leadership: Led a **multi-million-dollar cloud migration project** for a global enterprise, successfully transitioning legacy systems to **AWS and Azure** within the timeline and reducing operational costs by **25%**.

Automation Excellence: Implemented end-to-end **automation frameworks** for repetitive business processes, resulting in a **40% efficiency boost** and **15% cost savings** annually for a Fortune 500 client.

Global Consulting Impact: Directed strategic **AI and automation roadmaps** for clients in the **USA, UK, and Europe**, driving revenue growth by introducing **scalable, cloud-native AI solutions** that aligned with business objectives.

Certifications and Expertise: Achieved **GSDC Generative AI Professional Certification** and **AWS Solutions Architect Professional**, contributing to winning high-profile client engagements by ensuring technical excellence.

Cross-functional Leadership: Successfully managed **diverse, global teams** of over 50 members across engineering, product, and operations, delivering complex projects with **100% on-time delivery**.

AI-driven Insights: Developed **predictive analytics models** using Generative AI and ML, enabling a retail client to optimize inventory and reduce wastage by **20%**, saving over **\$2M annually**.

10/2010 – present
Bengaluru, India

AI Practice Head TATA CONSULTANCY SERVICES

AI Practice Head – Healthcare, Life Sciences, Energy, Resources & Utilities (HLERU)

- Spearheading the AI Practice for TCS's largest global business unit, covering Healthcare, Life Sciences, Energy, Resources, and Utilities with a focus on enterprise-scale AI adoption and transformation.
- Defined and executed the GenAI and Agentic AI strategy across industries, enabling clients to achieve operational efficiency, innovation, and compliance in highly regulated environments.
- Built and scaled a global AI practice team, driving talent development, solution accelerators, and reusable IP creation to strengthen TCS's AI portfolio.
- Led applied AI R&D initiatives, incubating solutions in clinical decision support, drug discovery, sustainable energy optimization, predictive maintenance, and intelligent grid operations.
- Architected and delivered enterprise-grade AI solutions including LLM-driven workflows, Retrieval-Augmented Generation (RAG) systems, and AI-powered SDLC automation frameworks.
- Recognised as a thought leader, representing TCS in CxO-level client engagements, industry forums, and analyst discussions on AI-led business transformation.

Key Use Cases:

- **Productivity Analytics:** Developed a **Generative AI-based tool** to capture and analyze **Copilot metrics usage**, providing insights into productivity improvement across teams.
- **SLLMs for Architecture Modernization:** Designed **GenAI-powered SLLMs** to eliminate **MSA (Microservices Architecture)** dependencies for a legacy application, improving scalability and maintainability.
- **Healthcare Automation:** Deployed an **AI-powered document analysis system** for clinical data, reducing processing time by **40%** and enhancing data accuracy.
- **Retail Optimization:** Built **predictive analytics models** for inventory management, reducing wastage by **20%** and saving millions in operational costs.
- **Customer Engagement:** Implemented **AI-driven chatbots** for patient support and retail customer queries, improving response time and user satisfaction by over **30%**.
- **AI for DevOps:** Developed **end-to-end automation frameworks**, integrating CI/CD pipelines with AI capabilities to accelerate software delivery by **60%**.
- Led the development and deployment of AI models using MLOps frameworks, ensuring seamless integration with CI/CD pipelines and real-time monitoring

Skilled in **cloud-native architectures** (AWS, Azure), cross-functional collaboration, and delivering scalable, impactful solutions that align with strategic business goals.

01/2008 – 06/2010
Bengaluru, India

Programmer Analyst Infosys Technologies

Cloud Automation Strategy: Designed and implemented **test automation strategies** tailored to cloud-native applications on **AWS**, ensuring alignment with project goals and comprehensive validation of functional and non-functional requirements, including performance and scalability.

Scripting and Framework Development: Developed and maintained **automation scripts** using **Python, Java**, and cloud-based testing tools, contributing to the enhancement of **automation frameworks** for improved scalability, maintainability, and seamless integration with **AWS services** (e.g., Lambda, S3, RDS).

Functional Automation : Collaborated with cross-functional teams to develop **test plans** for cloud-based solutions, identifying automation opportunities and ensuring thorough validation of **AWS deployments**. Automated test execution for **CI/CD**

Work Experience

pipelines, analyzing results and generating actionable insights for stakeholders. **Tool Selection and Integration:** Assessed and integrated industry-standard tools (e.g., **Terraform**, **Selenium**, **Appium**) and AWS-native services into testing workflows to enhance automation efficiency and ensure the reliability of results. Optimized **AWS architecture testing** to validate security, compliance, and scalability. **AWS Functional Validation:** Conducted detailed validation of cloud-based functionalities, including API integrations, serverless architectures, and storage solutions, ensuring **resilient and robust application performance** in a cloud environment.

11/2006 – 01/2008
Bengaluru, India

Automation Consultant Cognizant Technologies

Client Engagement: Engaged with clients worldwide to understand their unique business needs, challenges, and testing requirements. Provided tailored test automation consultancy services to enhance their testing processes.

Technology Expertise: Demonstrated proficiency in working with diverse technologies, platforms, and application architectures. Adapted quickly to new technologies and provided strategic guidance on the selection of appropriate automation tools and frameworks.

Global Project Leadership: Led international teams in implementing test automation solutions for clients in various industries. Collaborated with on-site and remote teams to ensure seamless communication, project delivery, and client satisfaction.

Automation Strategy Development: Developed comprehensive test automation strategies aligned with clients' business goals. Conducted feasibility studies, risk assessments.

08/2003 – 10/2006
Bengaluru, India

Analyst - Automation (Embedded) Sway Technologies

Define and execute the automation strategy, including selecting appropriate tools and frameworks. Identify opportunities for test automation and assess the feasibility of automation for different components. Write, review, and maintain automated test scripts using scripting or programming languages (e.g., Selenium, Java, Python). Ensure that automated scripts are efficient, maintainable, and reusable.

Execute automated test scripts and analyze test results. Investigate and troubleshoot test failures, identifying root causes and providing feedback to development teams. Collaborate with developers, business analysts, and other team members to understand product functionality and identify automation opportunities.

Participate in code reviews and contribute to overall product quality. Evaluate and recommend automation tools and frameworks based on project requirements. Implement and maintain automated testing tools and infrastructure.

Skills

Generative models like GANs, VAEs, and Transformer-based models (e.g., GPT, BERT).

Professional

Embedding generation and retrieval using vector databases: Chroma, Pinecone, ElasticSearch.

Infrastructure as Code (IaC): Terraform, AWS CloudFormation.

Developing and enhancing automation frameworks: Selenium, Appium, PyTest.

AI-Powered Applications: Conversational agents, text summarizers, document analysis tools

Professional

Hands-on with AI libraries: TensorFlow, PyTorch, Transformers

Professional

Monitoring and logging with CloudWatch, AWS X-Ray.

Functional Automation for cloud-based applications.

Skills

Automated validation for AWS services and serverless workflows.

Automated validation for AWS services and serverless workflows

MLOps Frameworks: MLflow, SageMaker
Professional

Workflow Automation with Docker/Kubernetes
Professional

Certificates

GSDC Generative AI Professional
Global Skill Development Council

AWS Certified Solutions Architect - Professional Certification
Amazon

Education

Lucknow, India

Electronics | B-Tech
Azad Institute of Engineering & Technology

Strengths

Expertise in Generative AI and Automation # Proven Leadership in Cloud Architecture
Accomplishments in Business-Driven AI Solutions # Global Experience and Collaboration
Strong Automation and DevOps Skills # Analytical and Strategic Thinking
Effective Communication and Documentation # Multi-Industry Impact # Leadership # Collaboration
Client Focused # Strategic Thinking # Problem Solving # Innovation # Adaptability